

Machine Learning-Based Prediction of HER2 Status Using DCE-MRI Radiomic Features

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Introduction

The Rising Global Burden of Breast Cancer

In 2022,

2.3 million

new cases were found

670,000

deaths occurred

By 2050,

A projected

38% increase

in new cases

And an expected

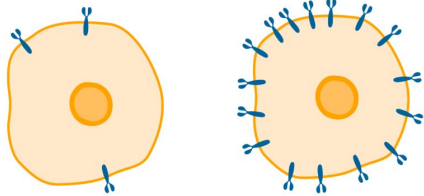
68% increase

in deaths

The growing burden of breast cancer highlights the need for improved tumor characterization.

HER2 Status is a Key Biomarker in Breast Cancer Care

Definition



Normal
Breast Cell

HER2-positive
Breast Cancer Cell

Treatment

- Guides treatment decisions
- HER2-targeted therapies
 - Trastuzumab
 - TKIs
 - ADCs
- Highly responsive to targeted therapies (receptor blocking)

Diagnosis

Tumor Detection (MRI, Mammography, Ultrasound)



Biopsy or Surgical Resection



Immunohistochemistry (IHC) test or fluorescence in situ hybridization (FISH) test

Subtypes	ER	PR	HER2
Luminal A	+	+/-	-
Luminal B	+	+/-	+/-
HER2+	-	-	+
TNBC	-	-	-

Medical Imaging and Machine Learning

Existing studies have demonstrated the successful application of machine learning (ML) to medical imaging data



Review

Machine Learning in MRI Brain Imaging: A Review of Methods, Challenges, and Future Directions

Martyna Ottoni ^{1,2,*}, Ann npj | digital medicine

www.nature.com/npjdigitalmed

REVIEW ARTICLE OPEN

Machine learning for medical imaging: methodological failures and recommendations for the future

Gaël Varoquaux^{1,2,3,5,6} and Veronika Cheplygina^{6,5,8}



Deep learning techniques in PET/CT imaging: A comprehensive review from sinogram to image space

Maryam Fallahpoor^a, Subrata Chakraborty^{a,b}, Biswajeet Pradhan^{a,c,*}, Oliver Faust^d, Prabal Datta Barua^{b,e,f}, Hossein Chegeni^g, Rajendra Acharya^h



MRI	DCE-MRI
• Anatomy • Single image • Structural features	• Anatomy + Function • Time-series data • Structural + kinetic features

- ❑ Especially insightful for certain tissues (**breast**, prostate, liver)
- ❑ Relatively underexplored compared to conventional MRI
- ❑ Rich data source for machine learning models

Research Motivation and Objectives

Research Gap

- Smaller patient cohorts in prior studies
- Limited research on DCE-MRI and ML models
- Limited comparison of machine learning models
- Unclear which feature groups best predict HER2 status

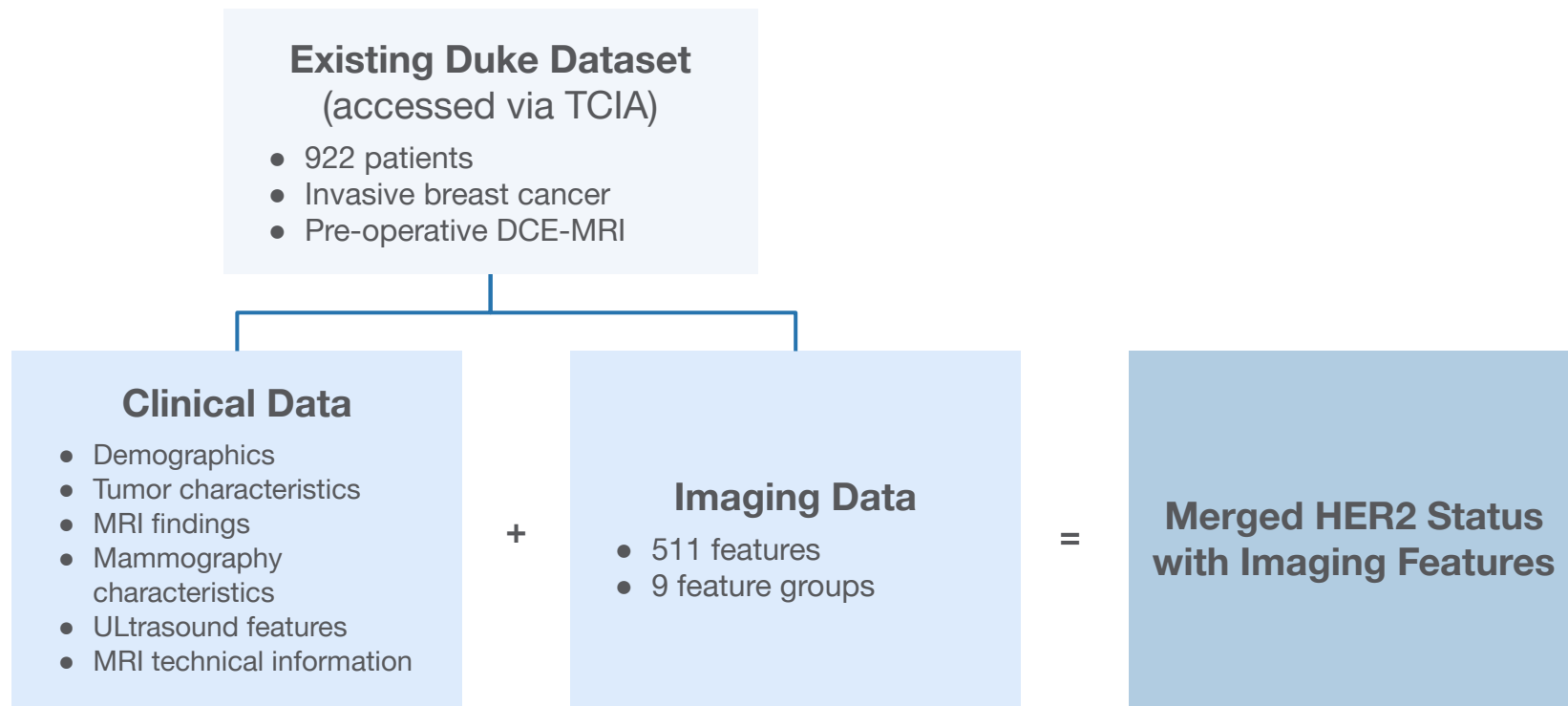


Research Objectives

- 1) Classify HER2 status using DCE-MRI radiomic features
- 2) Identify the most informative feature groups
- 3) Compare Logistic Regression, Random Forest, and SVM performance

Methods

Study Cohort and Data Access



Data Preprocessing

Merging Data

- Matching clinical patient data with images using unique identifiers
- Predictors: Imaging features (x)
- Outcome: HER2 status (y)

Missing Data

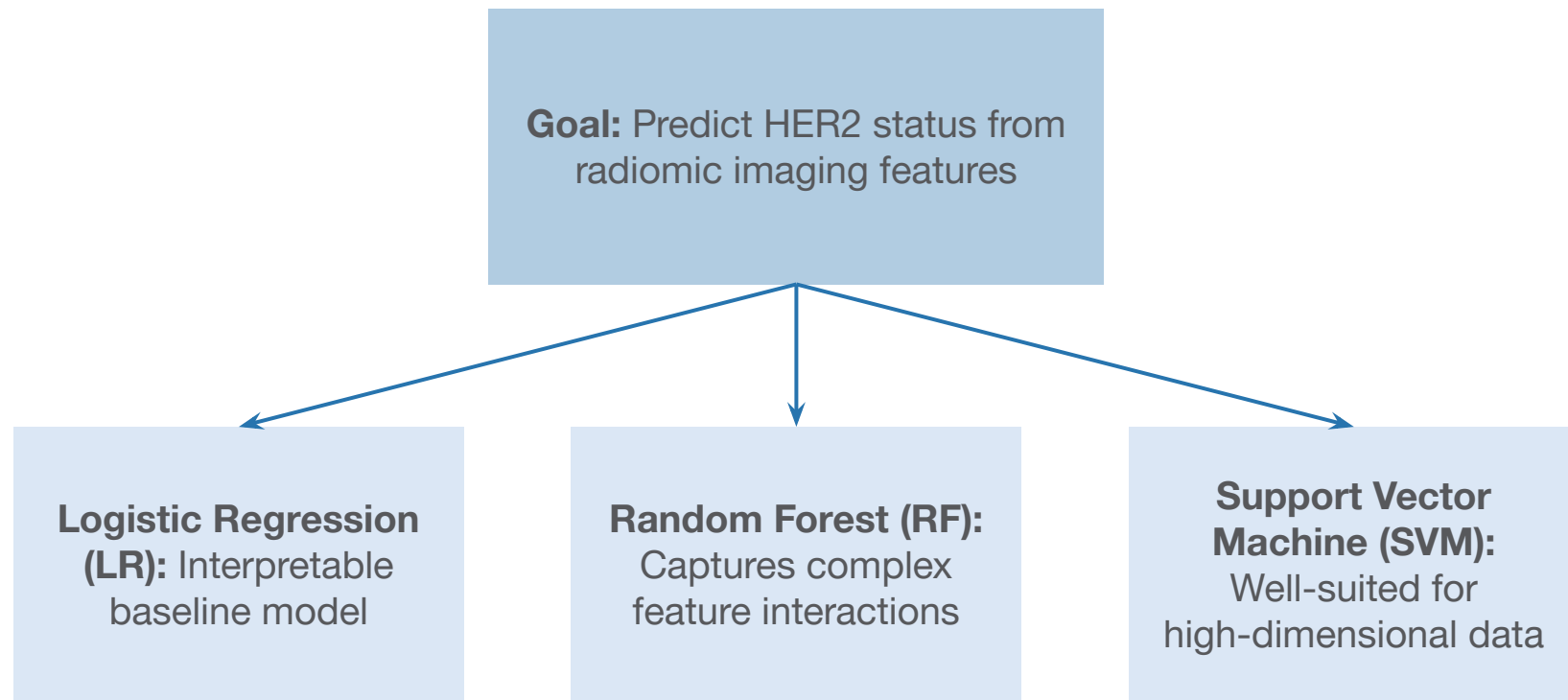
- Very few missing entries
 - 0.11% of all imaging features
 - Highest column missingness 4.77%
- Median data imputation

Train/Test Split

- 80/20 split
 - Training, $n = 737$
 - Testing, $n = 185$
- Preserved HER2+ class distribution via stratification



Three Machine Learning Models Were Evaluated



Results

HER2 Classification Performance Across Feature Groups

What does AUC measure? AUC measures the model's ability to distinguish between HER2+ and HER2- tumors. Higher values indicate better classification performance.

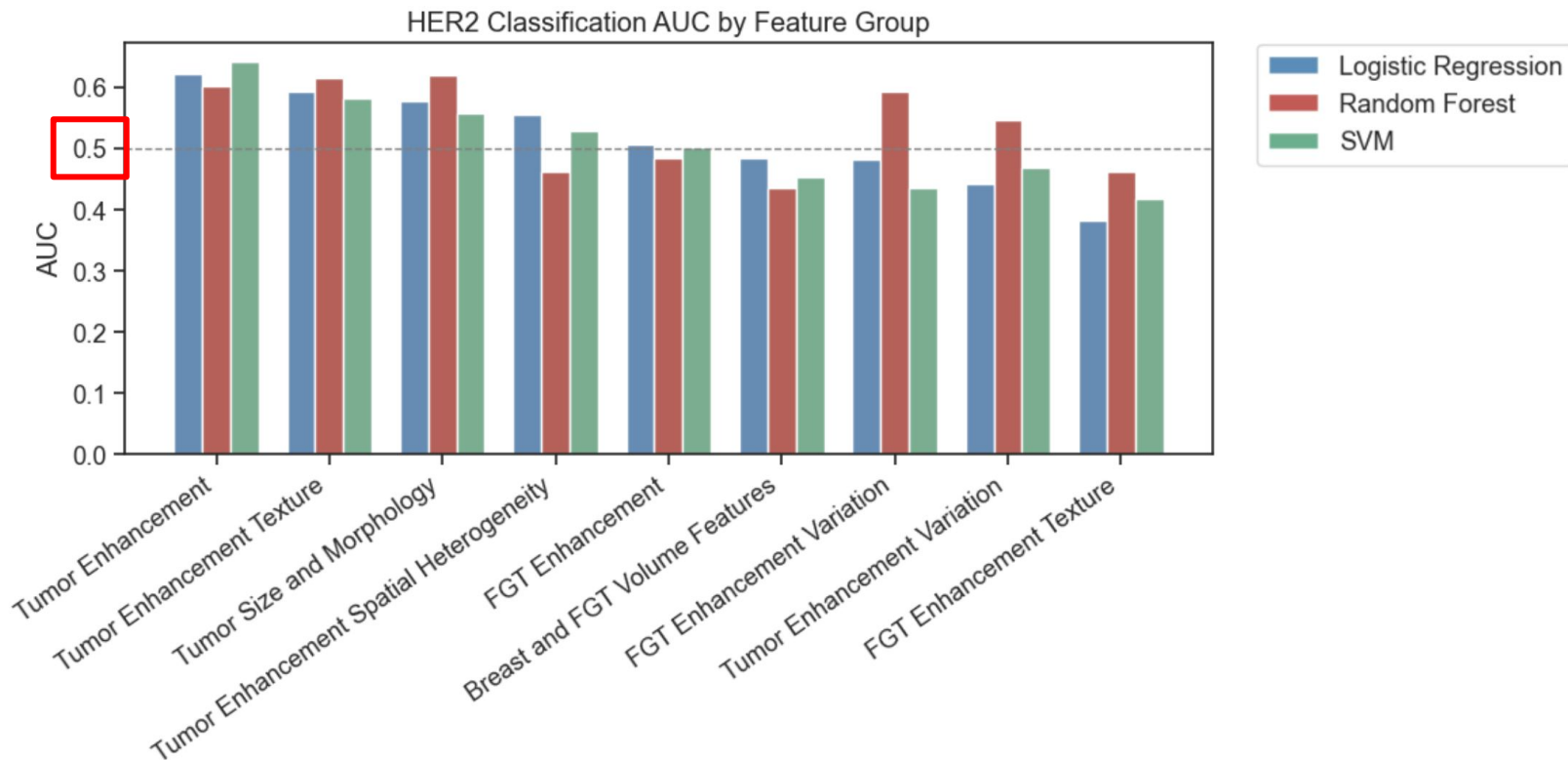
Table 2. Area Under the ROC Curve (AUC) for HER2 Status Prediction Across Feature Groups and Machine Learning Models				
Feature Group	N Features	LR	RF	SVM
Breast and FGT volume features	5	0.484	0.434	0.453
Tumor size and morphology	10	0.576	0.618	0.556
FGT enhancement	82	0.505	0.483	0.502
Tumor enhancement variation	31	0.442	0.593	0.468
FGT enhancement texture	176	0.381	0.462	0.418
Tumor enhancement texture	135	0.592	0.614	0.581
Tumor enhancement spatial heterogeneity	4	0.555	0.462	0.528
FGT enhancement variation	34	0.481	0.593	0.434
Tumor enhancement	30	0.621	0.546	0.641
LR = Logistic Regression, RF = Random Forest, SVM = Support Vector Machine				

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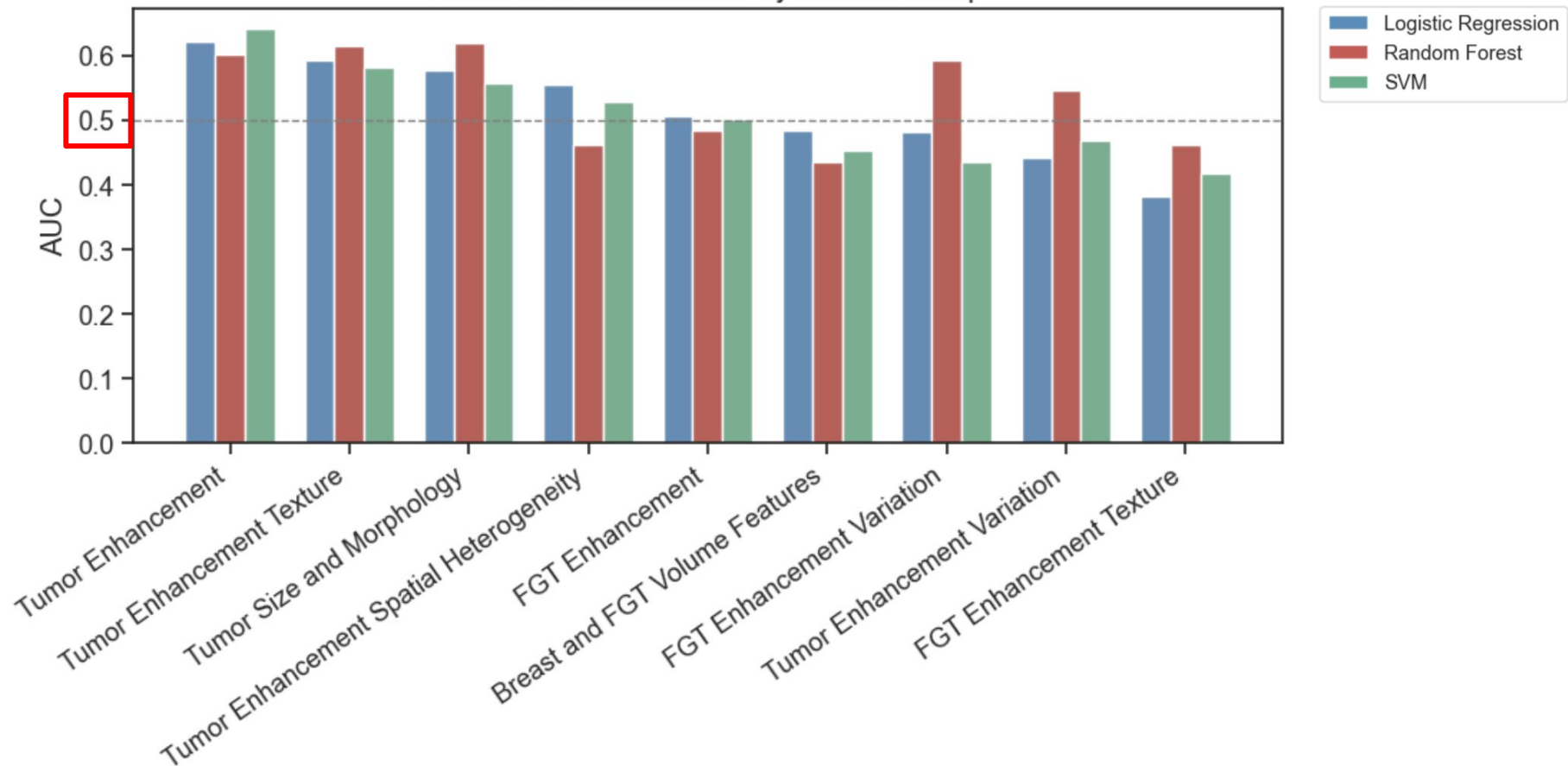
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Comparing Classification Performance Across Feature Groups

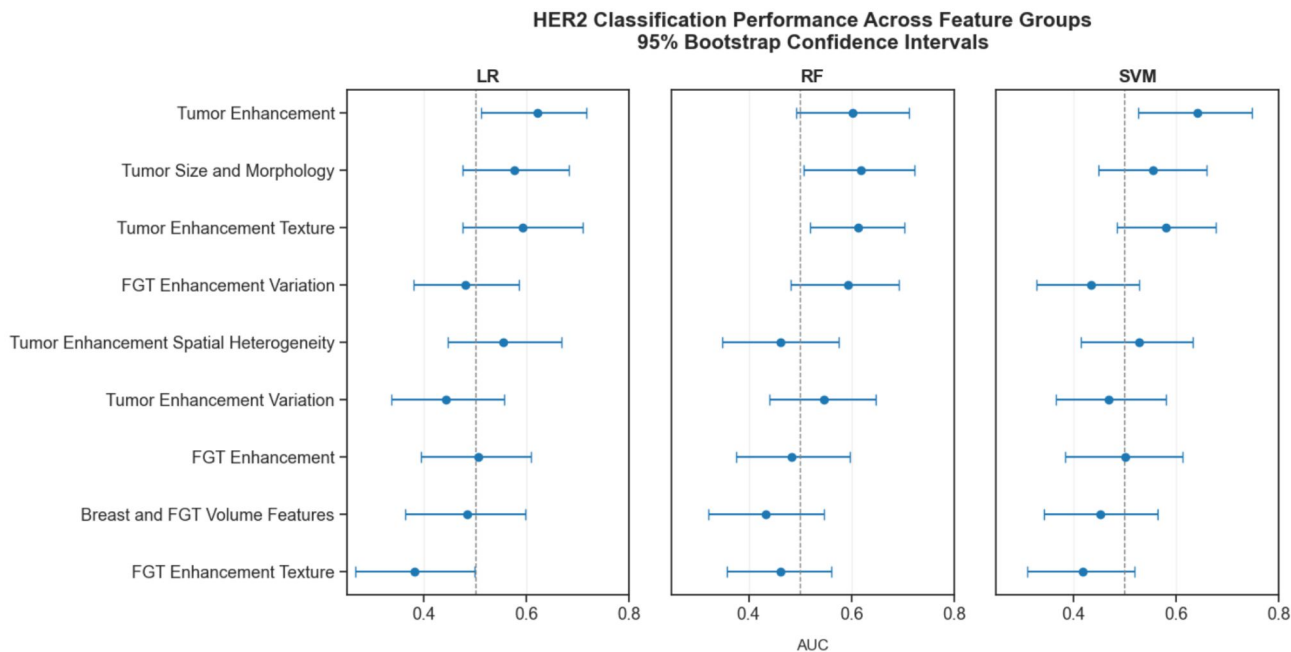


HER2 Classification AUC by Feature Group

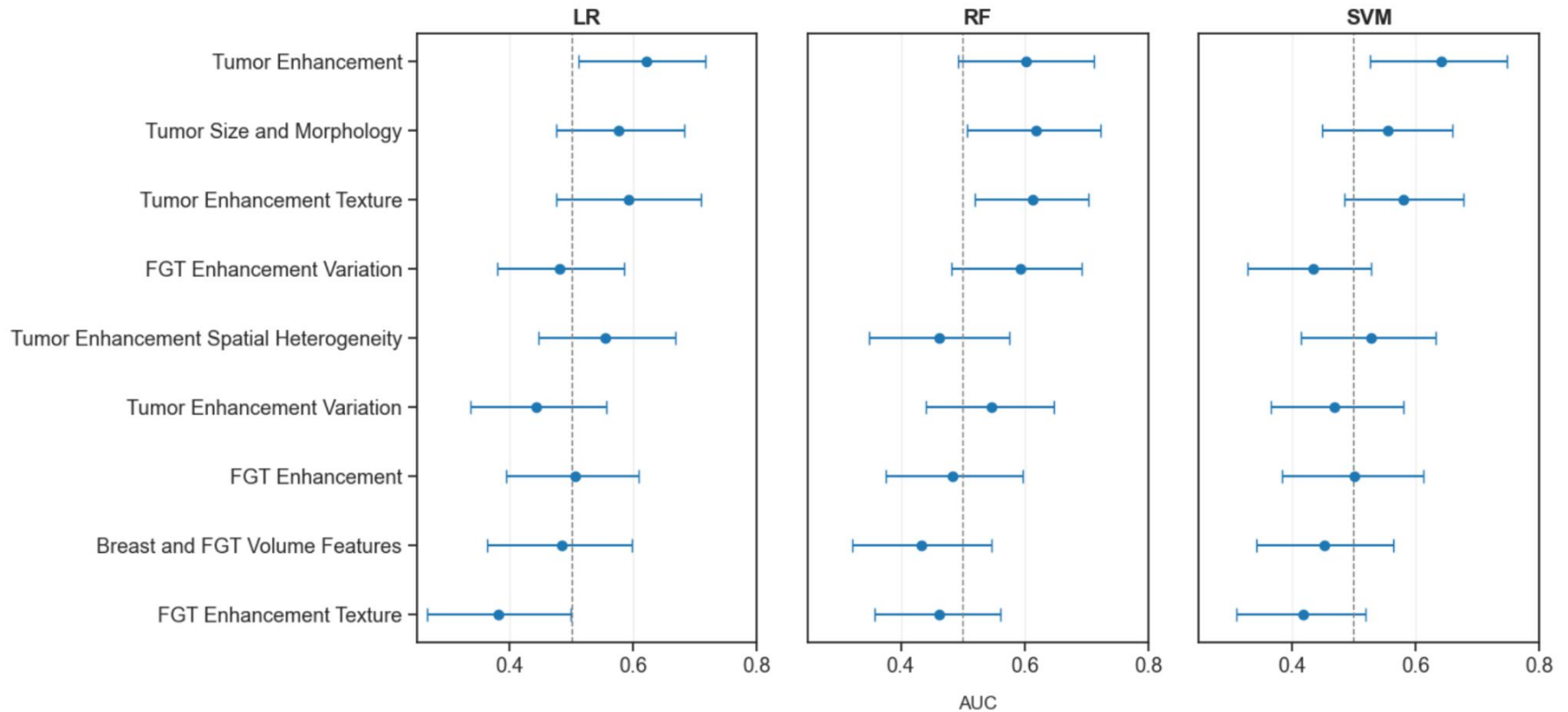


Confidence Intervals for HER2 Classification Performance

A point estimate alone does not capture uncertainty in model performance, so 95% bootstrap confidence intervals were generated using 1,000 resamples of the test set.



HER2 Classification Performance Across Feature Groups 95% Bootstrap Confidence Intervals



Discussion

Key Findings and Implications

1 Tumor feature groups tended to perform best. Tumor Enhancement, Tumor Enhancement Texture, Tumor Size and Morphology consistently had the highest AUCs.

2 Predictive performance remained limited overall. No model exceeded an AUC of 0.65.

3 Models produced similar results. While RF achieved the highest AUCs for several feature groups, performance differences between models were relatively small.

Future Extensions of this Project

Short term:

- Hyperparameter tuning
- Additional machine learning models
- Alternative train/test splits
 - Ex: Saha et al. 2018 utilized a 50/50 split
- Cross-validation

Long term:

- Feature-level analysis within top-performing groups to identify the most predictive radiomic features
- Integration of clinical and imaging features to make stronger predictions
- Extension to multi-class or continuous clinical outcome variables

Thank You
